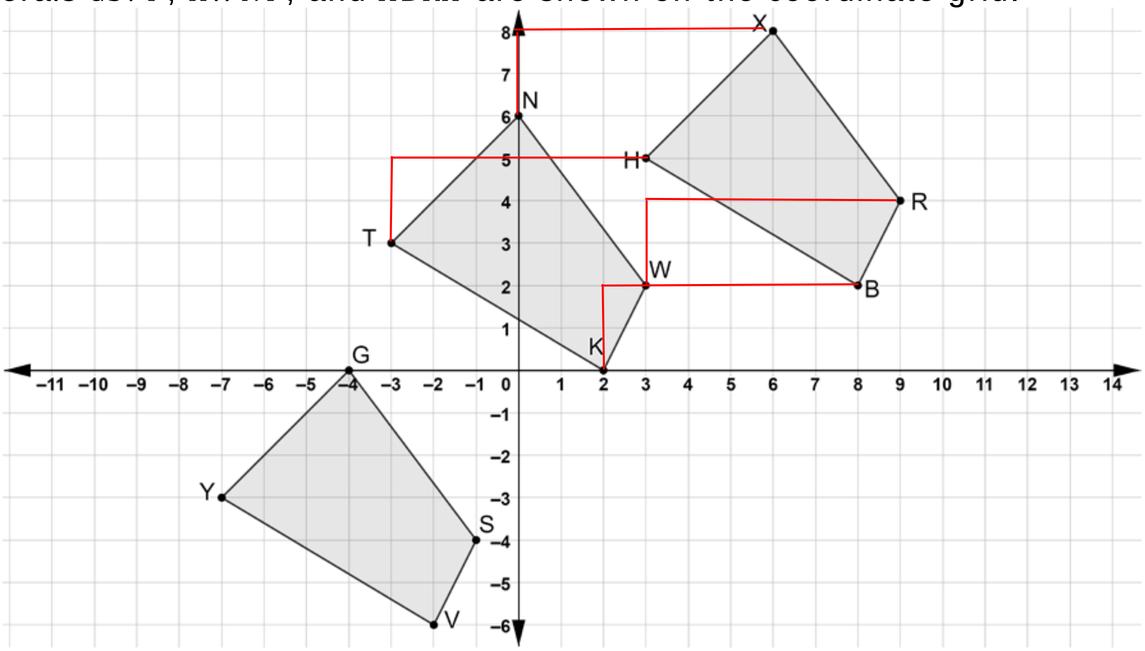
Geometry Name_Answer Key_________
Unit 4 Date_____________________
Lesson 1 Homework Period___________________

Quadrilaterals $GSVY$, $KWNT$, and $HBRX$ are shown on the coordinate grid.



1. Draw the mapping that shows the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.

2. Describe the translation if quadrilateral $TKWN$ is the preimage and quadrilateral $HBRX$ is the image.

   up 2, right 6

3. Complete the statements.
   The x-values of quadrilateral $GSVY$, the image, can be found by

   - o adding
   - ★ subtracting
   - o multiplying

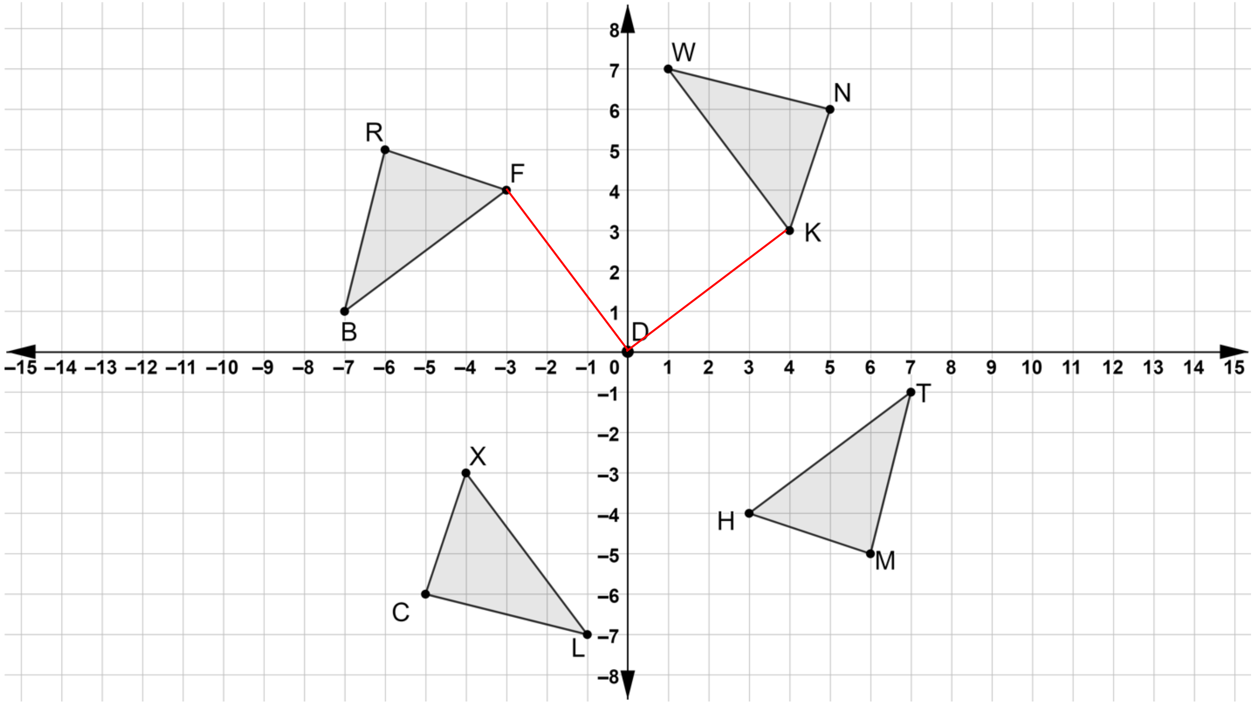
   _10_ to the x-values of quadrilateral $XRBH$, the pre-image. The y-values of quadrilateral $GSVY$, the image, can be found by

   - o adding
   - ★ subtracting

   _8_ units to the y-values of quadrilateral $XRBH$, the pre-image.

4. Describe the translation if quadrilateral $XRBH$ is the preimage and quadrilateral $GSVY$ is the image.
   left 10, down 8

Four triangles are shown on the coordinate grid. Triangles WNK, TMH, and CXL are all rotations of $\triangle RFB$.



5. On the coordinate grid draw angle FDK.

6. Describe how the measure of $\angle FDK$ can be used to determine the degrees of rotation where $\triangle RFB$ is the pre-image and $\triangle NKW$ is the image.

The angle measurement is the angle of rotation when the center of rotation is the origin.

7. Describe how the coordinates of the vertices of $\triangle RFB$, the preimage, and the coordinates of $\triangle NKW$, the image can be used to determine the degrees of rotation.
How the coordinates change can indicate the angle of rotation.

8. Describe the rotation where $\triangle RFB$ is the preimage and $\triangle CXL$ is the image.

One quadrant, 90° counterclockwise.

9. Describe how the coordinates of the vertices of $\triangle RFB$ and $\triangle TMH$ show that rotating $\triangle RFB$ $180°$ about the origin results in $\triangle MHT$.

$R(-6,5)$	$T(7,-1)$	Both x and y coordinates change sign between preimage and image.
$F(-3,4)$	$M(6,-5)$	
$B(-7,1)$	$H(3,-4)$	

10. Explain why the rotation of $\triangle RFB$ that results in $\triangle MHT$ can be described as $180°$ clockwise or $180°$ counterclockwise.

Moving 180° counterclockwise is the same as moving 180° clockwise.